Lecture #13: The Simplex Method

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A Refresher- Standard Form

Definition (Standard Form)

The *standard form* of a linear programming problem is

$$\begin{aligned} \text{minimize } z = \mathbf{c}^\top \mathbf{x} \text{ subject to} \\ \mathbf{A}\mathbf{x} \le \mathbf{b} \\ \mathbf{x} \ge \mathbf{0}, \end{aligned}$$

where $\mathbf{A} = [a_{ij}]$ is an $m \times n$ matrix,
$\mathbf{b} = \begin{bmatrix} b_1 & b_2 & \cdots & b_m \end{bmatrix}^\top$, $\mathbf{c} = \begin{bmatrix} c_1 & c_2 & \cdots & c_n \end{bmatrix}^\top$, and $\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}^\top$

We need to make the assumption that $\mathbf{b} \ge \mathbf{0}$.
We now turn the standard form into the canonical form, by adding in slack variables $x_{n+1}, \cdots x_{n+m}$.

A Refresher- Canonical Form

Definition (Canonical Form)

$$\begin{aligned} \text{minimize } z = \mathbf{c}^\top \mathbf{x} \text{ subject to} \\ \mathbf{A}\mathbf{x} = \mathbf{b} \\ \mathbf{x} \geq \mathbf{0}, \end{aligned}$$

where $\mathbf{A} = [a_{ij}]$ is the $m \times (n+m)$ matrix,

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & 1 & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & a_{2n} & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & 0 & 0 & \cdots & 1 \end{bmatrix}$$

and $\mathbf{c} = \begin{bmatrix} c_1 & c_2 & \cdots & c_n & 0 & \cdots & 0 \end{bmatrix}^\top$, and $\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n & x_{n+1} & \cdots & x_{n+m} \end{bmatrix}^\top$

The Simplex Algorithm, simply

Definition (Adjacent points)

Two distinct extreme points in S are said to be *adjacent* if as basic feasible solutions they have

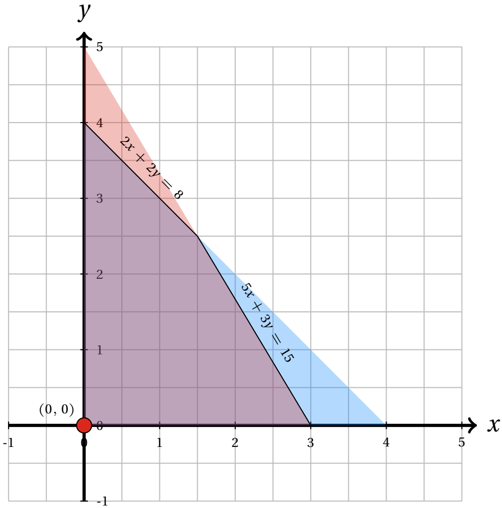
Remark

The idea behind the simplex algorithm is to start at a point, go to adjacent points that we know will further optimize the objective function and repeat until we know we have the optimal value. To find the adjacent points, we bring a new variable in ("entering variable") and dismiss another ("departing variable"). We can only have m basic variables; here $m = 2$.

Setting up the Simplex Method

$$\text{maximize } z = 120x + 100y \text{ subject to}$$
$$2x + 2y \leq 8$$
$$5x + 3y \leq 15$$
$$x, y > 0$$

First, we need to convert to canonical form.



y
x
5
4
3
2
1
0
-1
-1
0
1
2
3
4
5
$2x + 2y = 8$
$5x + 3y = 15$
(0, 0)

Setting up the initial tableau

Now that we have all of the equations as equalities, we want to construct the initial tableau.

In the tableau, a basic variable (right now u, v) has the following properties:

1. It appears in exactly one equation and in that equation it has a coefficient of +1
2. The column that it labels has all zeros (including the objective row entry) except for the +1 in the row that is labeled by the basic variable
3. The value of a basic variable is the entry in the same row in the rightmost column.

Definition (Optimality Condition)

If the objective row of a tableau has zero entries in the columns labeled by basic variables and no negative entries in the columns labeled by nonbasic variables, then the solution represented by the tableau is optimal.

	x	y	u	v	z	b
u	2	2	1	0	0	8
v	5	3	0	1	0	15
	-120	-100	0	0	1	0

Identifying the Entering Variable

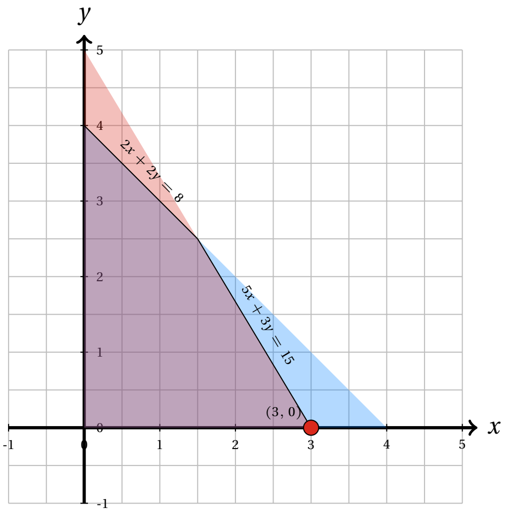
- Since we are in canonical form, to increase the value, we must both increase a nonbasic variable, and decrease a basic varaible
- The largest increase in z per unit increase in a variable occurs for the most negative entry in the objective row.
- This occurs under the x column, so that x is chosen to be the variable to be increased from zero to a positive value.

Identifying the Departing Variable

Since we are only allowed to have two basic variables, we need to select one to leave as well.

Identifying the Departing Variable (cont.)

Visualizing



Definition (Departing Variable)

We now have the variable $v = 0$. It is no longer a basic variable because it is zero, and it is called a *departing variable* since it has departed from the set of basic variables.

Definition (Pivotal Row, Pivotal Column)

The column of the entering variable is called the *pivotal column*; the row that is labeled with the departing variable is called the *pivotal row*.

Example (What if we picked the other ratio?)

We had $x \leq 8/2$ or $x \leq 15/5$, This means that the next basic slution is not feasible.

After we have figured out which have entered and departed, we need to make a new tableau. Solving the second constraint, we have

A more Matrix Way to think about this

1. Locate and circle the entry at the intersection of the pivotal row and pivotal column. This entry is called the pivot. Mark the pivotal column by placing an arrow $\downarrow$ above the entering variable, and mark the pivotal row by placing an arrow $\leftarrow$ to the left of the departing variable.
2. If the pivot is k, multiply the pivotal row by $1/k$, making the entry in the pivot position in the new tableau equal to 1.
3. Add suitable multiples of the new pivotal row to all other rows (including the objective row), so that all other elements in the pivotal column become zero.
4. In the new tableau, replace the label on the pivotal row by the entering variable

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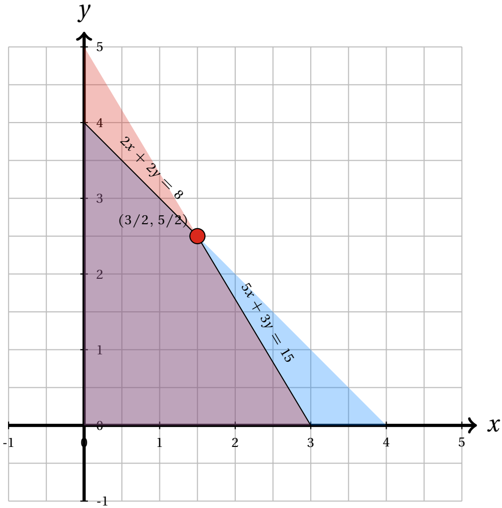
	x	y	u	v	z	b
u	0	$\frac{4}{5}$	1	$\frac{-2}{5}$	0	2
x	1	$\frac{3}{5}$	0	$\frac{1}{5}$	0	3
	0	-28	0	24	1	360

Are we done?

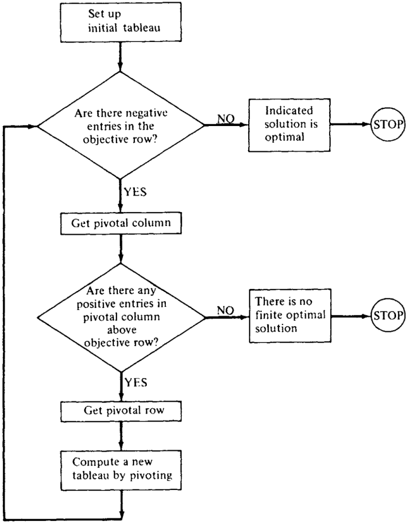
Since the pivot was $\frac{4}{5}$, we multiply everything by $\frac{5}{4}$ to get the pivotal row to be

Since there are no negative entries in the objective row, we are done, as we have met the optimality condition.



y
x
5
4
3
2
1
0
-1
-1
0
1
2
3
4
5
2x + 2y = 8
(3/2, 5/2)
5x + 3y = 15

A Simplex Flow Chart



Another Example

$$\text{maximize } z = 4x_1 + x_2 + 4x_3 \text{ subject to}$$

$$\begin{aligned} 2x_1 + x_2 + x_3 &\leq 2 \\ x_1 + 2x_2 + 3x_3 &\leq 4 \\ 2x_1 + 2x_2 + x_3 &\leq 8 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

First, we need to get it to canonical form

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Now we need to set up the initial tableau:

Are there negative entries in the objective row?
Are there any positive entries in the pivotal column above the objective row?
Compute the new tableau by pivoting.
We need to compute the pivot. The first row we have, $\frac{1}{1} = 1$ and the second row we have $\frac{4}{1} = 4$. Since we need to pick the smaller of the two, $\min\{1, 4\} = 1$, so the pivot is x_1 and u will be departing us.

Now we row reduce to get the new pivot to be a basic variable

Are there negative entries in the objective row?
Are there any positive entries in the pivotal column above the objective row?
Compute the new tableau by pivoting.
We need to compute the pivot. The first row we have, $\frac{1}{0.5} = 2$ and the second row we have $\frac{3}{2.5} = \frac{6}{5} = 1.2$. Since we need to pick the smaller of the two $\min 2, 1.2 = 1.2$, so the pivot is x_2 and v will be departing us.

Now we row reduce

Are there negative entries in the objective row?

Therefore, we have found the optimal solution: $x_1 = 0.4$, $x_2 = 0$, $x_3 = 1.2$, achieving the maximum value: $z = 6.4$.

Your Turn: (a great homework question style question)

Example

$$
\begin{aligned}
\text{maximize } z = 2x + 5y \text{ subject to} \\
3x + 5y &\le 8 \\
2x + 7y &\le 12 \\
x, y &\ge 0
\end{aligned}
$$